

To Face or Not To Face

Determining whether an image is a face or not

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Modeling Question

For our project we set out to use PCA to determine whether an image showed a face or not a face.

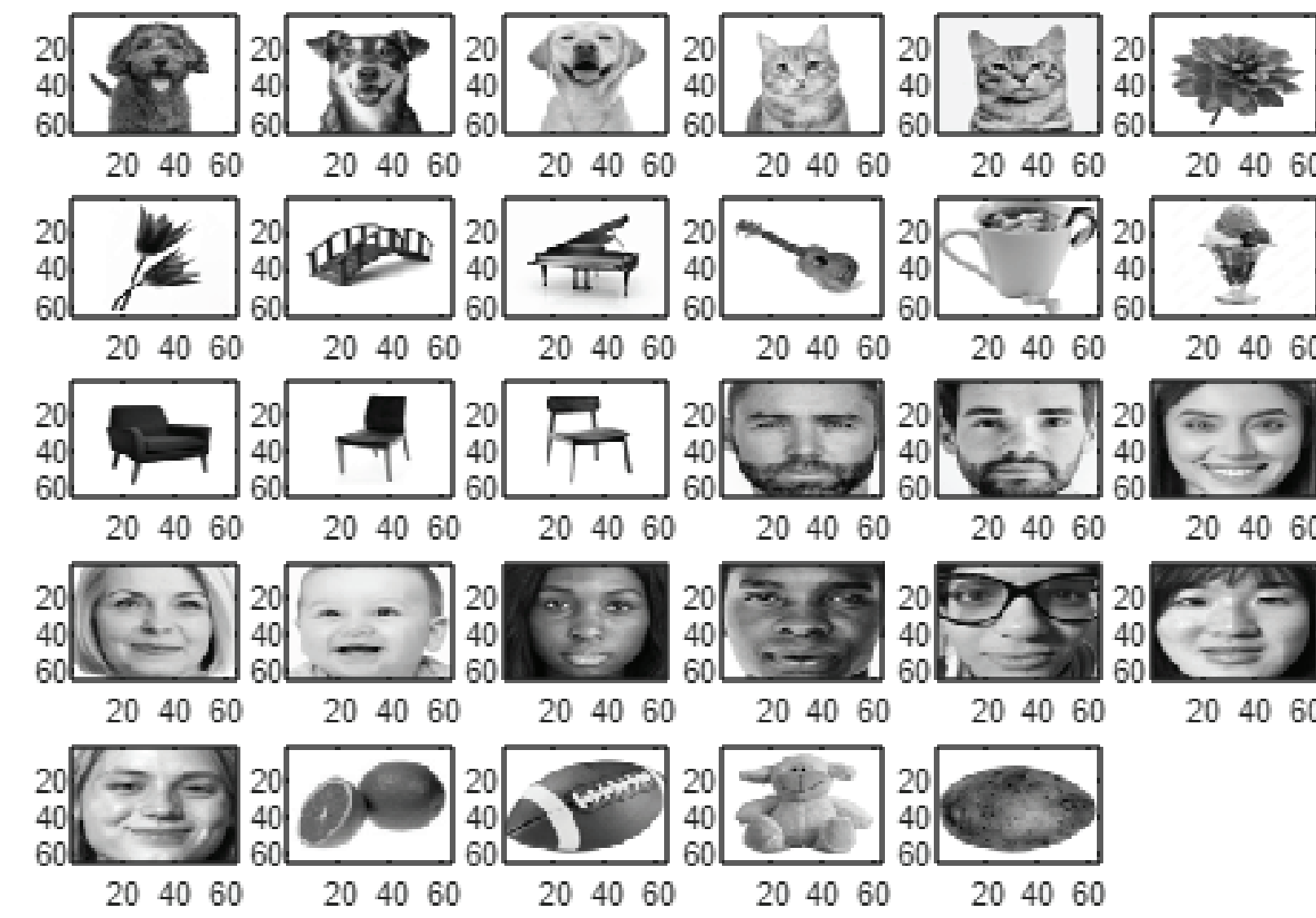
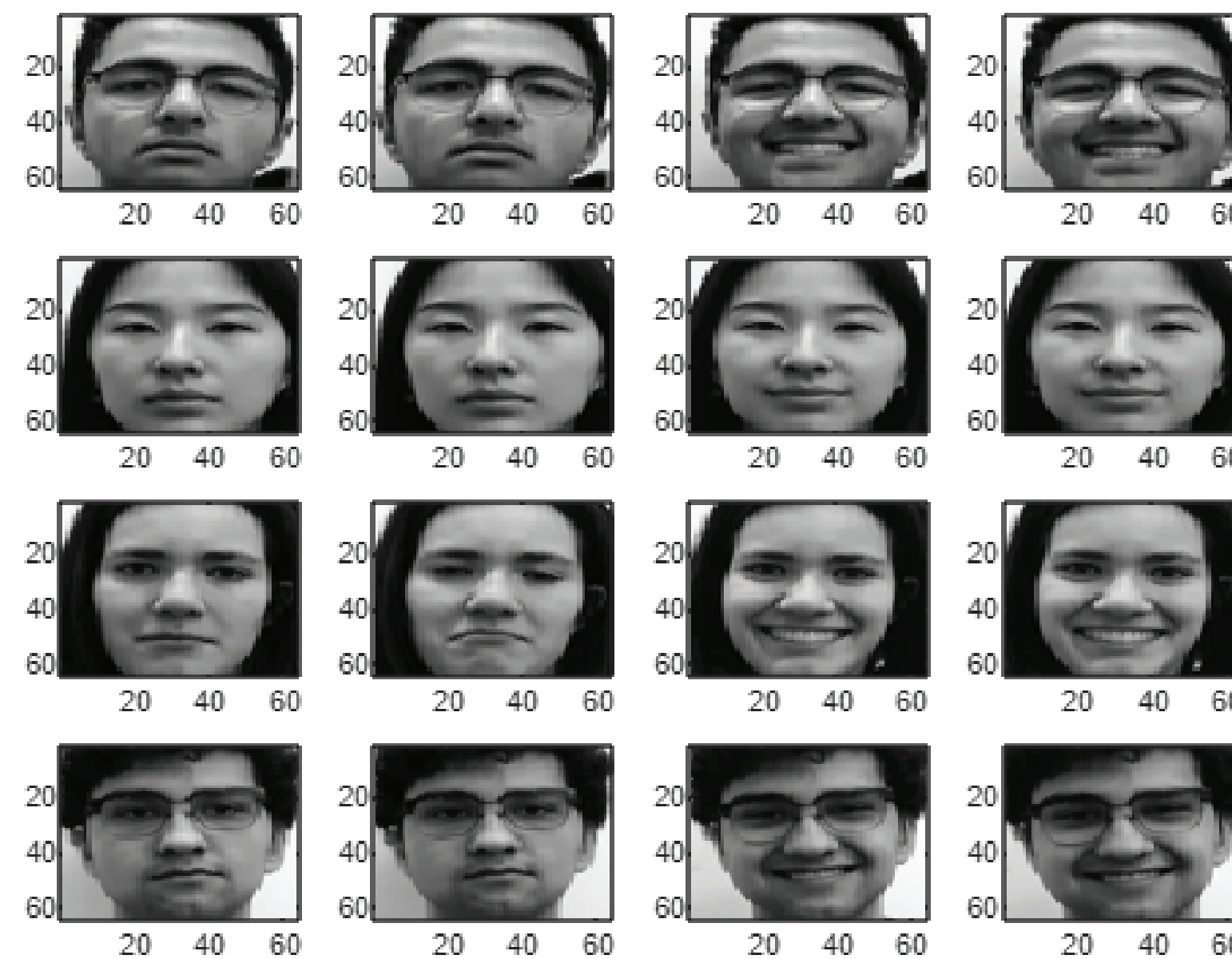
Which is a photo of a face?



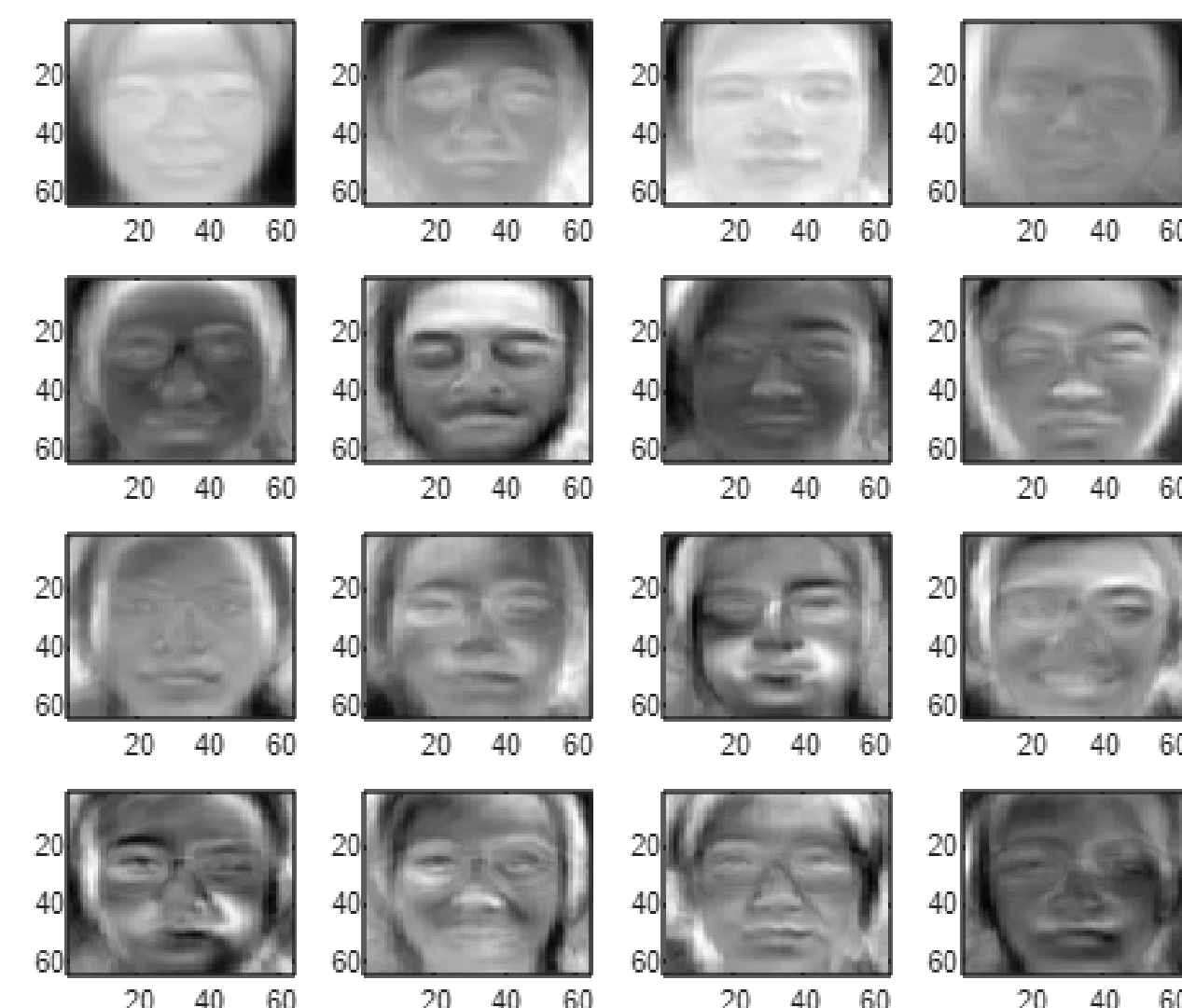
Making our Model

First we found our three data sets:

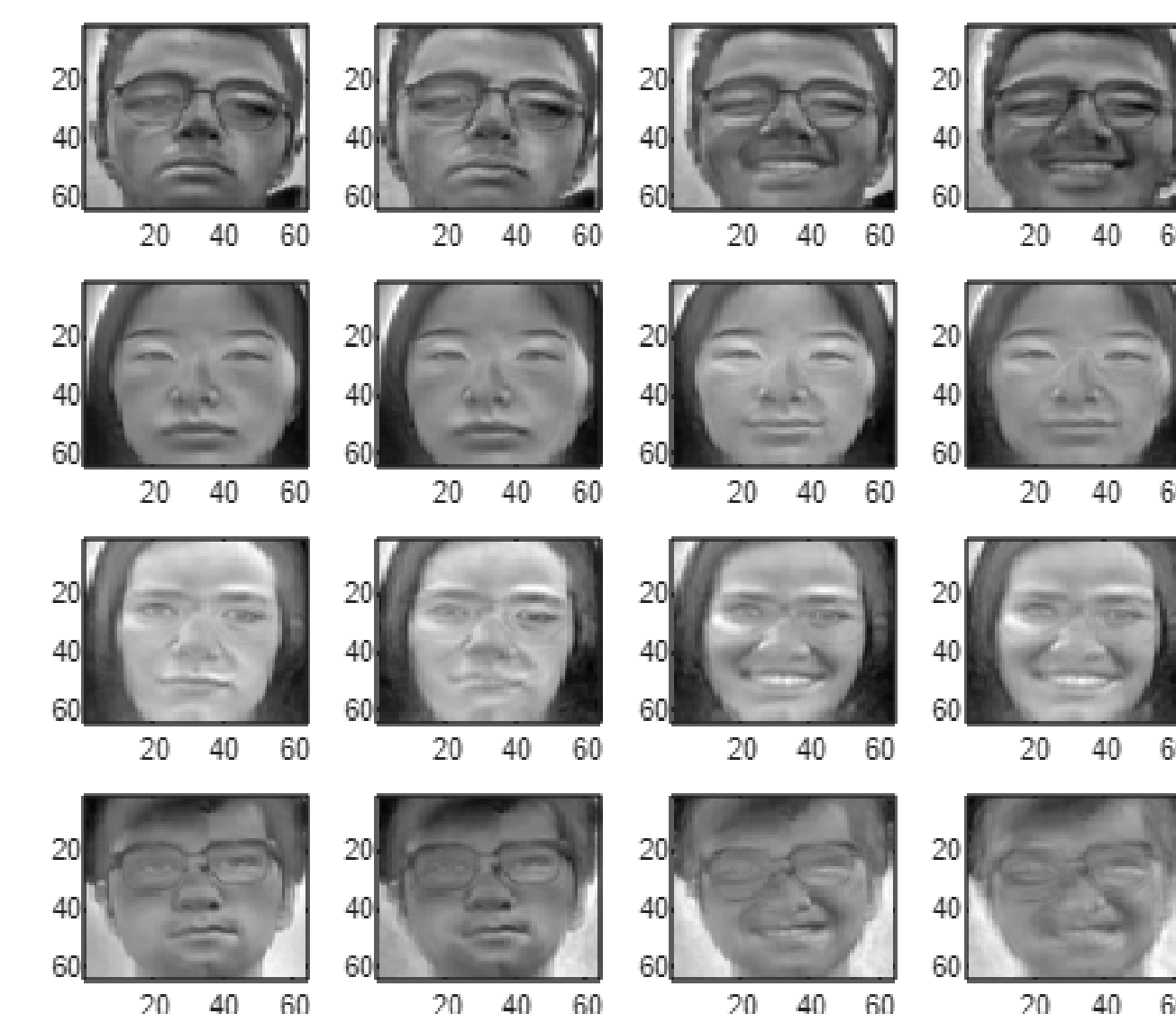
Set a, containing only photos of faces, set b, containing only photos of faces and set c containing a combination of faces as well as non-faces



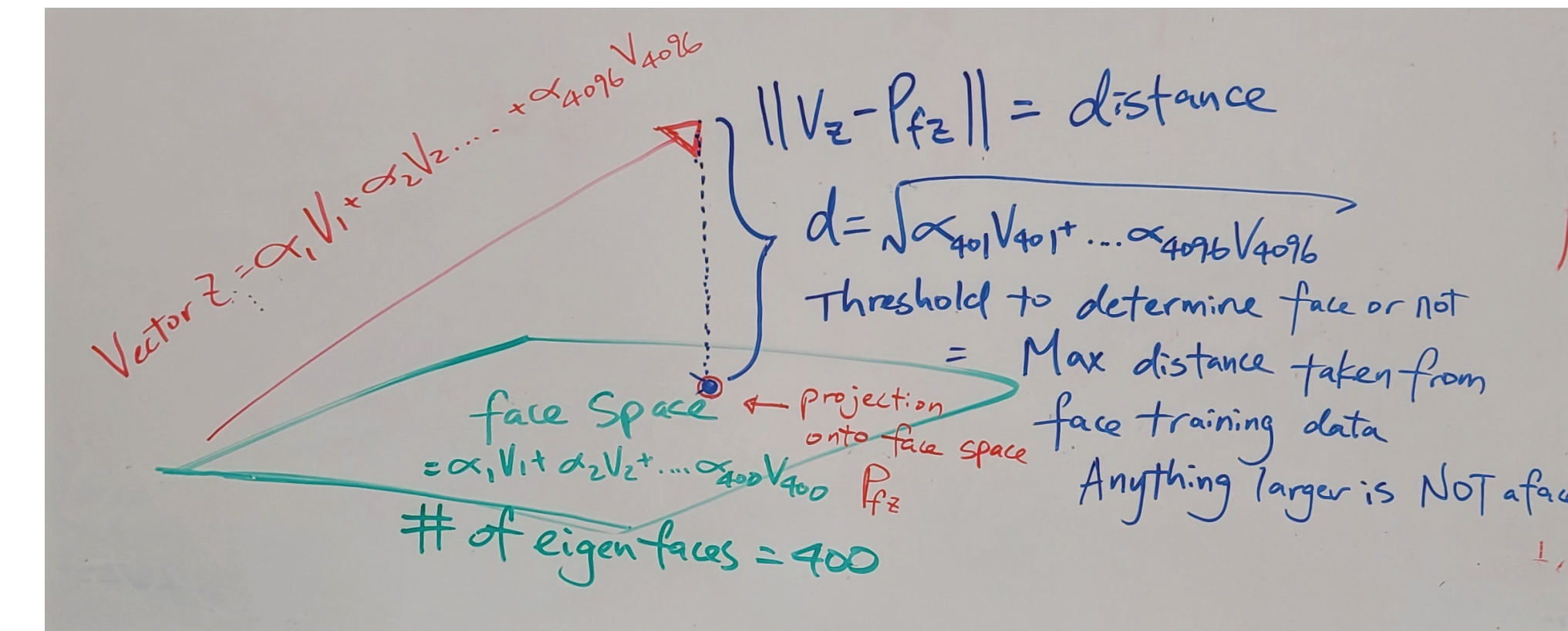
On our first set of training images we performed pca to find the eigenfaces that would make up our facespace. Here are 16 of our eigenfaces:



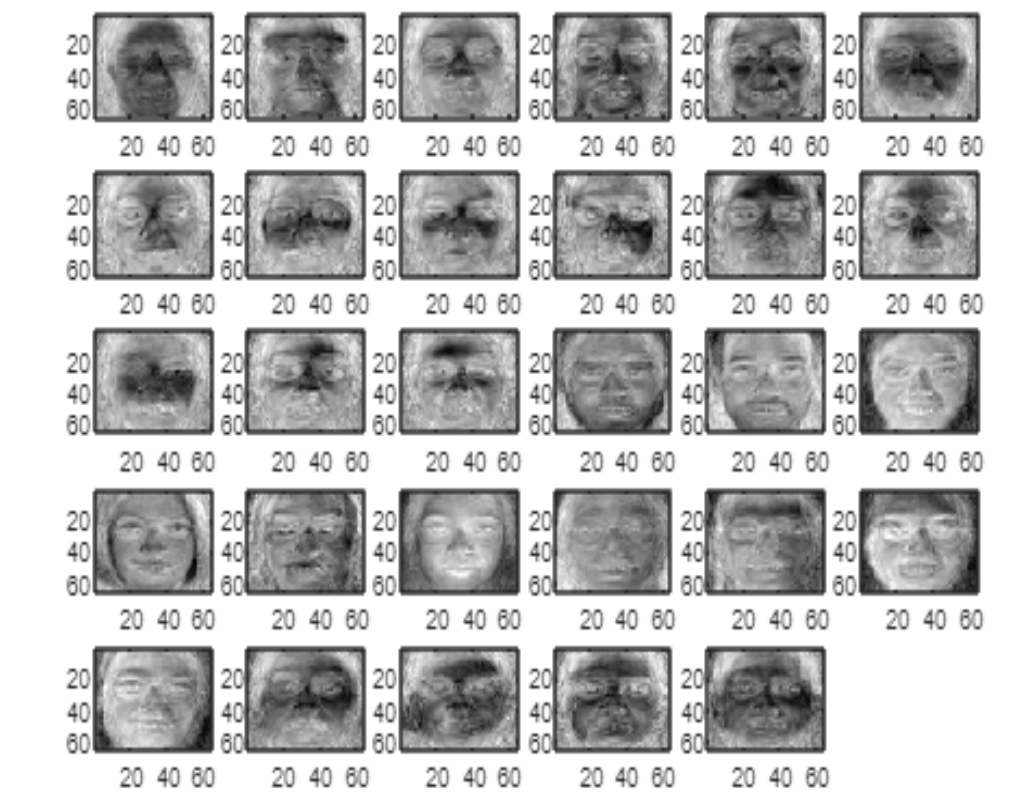
Next we projected the faces from our set b data set into the facespace of set a.



Then we found the distance between each of the projected faces from set c into set a to each of the non-projected faces in set c. Next we compared the distances from set c to the maximum distance for a face in set b. If the distance for the image was closer or equal to the maximum distance then the image was deemed a face.

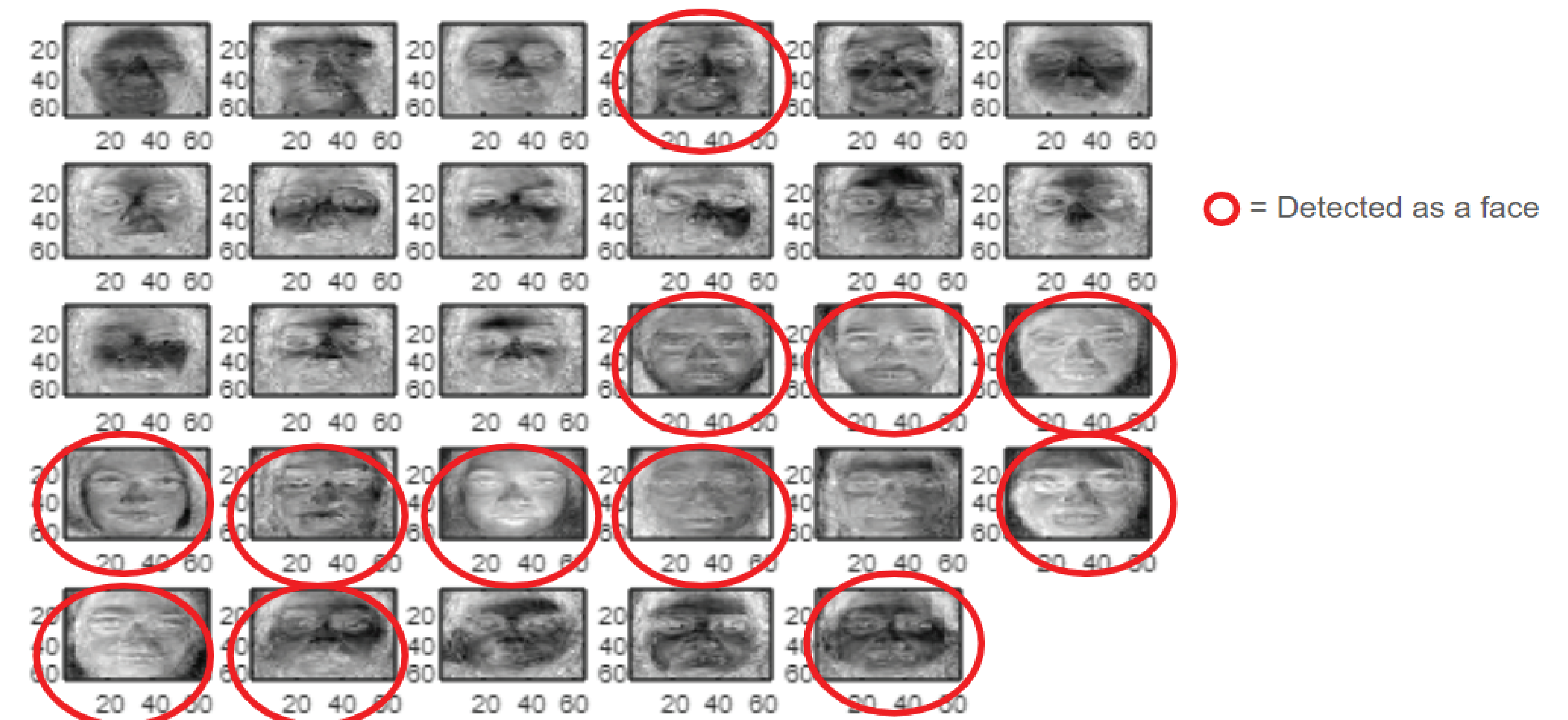


Next we projected our data from set c into the face space of set a



Results

Our algorithm detected 90% of the faces correctly as faces and detected 15.8% of the non-faces as faces



Future Work and Conclusions

For our future work on this project we would try to fix the problem of overfitting by adding more varied training images to our dataset. This way our project will only be fitted to 80% of the data, allowing it to be more accurate because it detects the more general shape of a face as opposed to only detecting faces that are extremely similar to the training data.

Limitations

- Only 356 training images
 - cannot capture full variability in human faces
 - creates overfitting to our specific dataset
- Code doesn't include face alignment nor preprocessing
 - cannot handle variations in lighting, pose, and facial expressions